

Chapter 5

Groups

A *binary operation* on a set is, roughly speaking, a way of combining pairs of elements from a set to produce a new element. We see many examples of binary operations in mathematics in the topics we have studied up to now. We see numbers, of various sorts, and the arithmetic operations of addition, multiplication etc that we can perform on them. We have met matrices and the operations of matrix addition and multiplication. We have met functions and the operations of addition, multiplication and composition. In these various systems we see a lot of similarities in how the operations work. These common properties of operations on sets are what forms the subject matter of *abstract algebra*. And one of the primary types of object studied in abstract algebra is a group.



5.1 Group axioms

Definition 5.1. A *group* $(G, *)$ is a pair consisting of a non-empty set G together with a binary operation $*$ defined on G such that the following axioms hold,

1. (*Closure*) $\forall a, b \in G \quad a * b \in G$,
2. (*Associativity*) $\forall a, b, c \in G \quad a * (b * c) = (a * b) * c$,
3. (*Existence of identity*) $\exists e \in G \quad \forall a \in G \quad e * a = a * e = a$,
4. (*Existence of inverses*) $\forall a \in G \quad \exists a^{-1} \in G \quad a * a^{-1} = a^{-1} * a = e$.

Note that the group is the pair consisting of the set and the operation that together satisfy the four group axioms. This is an important point as there may be more than one operation that one can define on a given set G such that it forms a group with respect to that operation. However in practice we may relax this formalism and simply refer to ‘the group G ’, as long as what the operation is is understood and no confusion will arise.

Definition 5.2. A group $(G, *)$ is *Abelian* if and only if the operation $*$ is commutative on G , i.e.

$$\forall a, b \in G \quad a * b = b * a.$$

The commutative property is not included as a group axiom because there are many important examples of systems that are not commutative, for example any systems of functions with the composition operation, or a system of matrices with the multiplication operation. Indeed, being Abelian is considered quite a special property of a group, and it simplifies much of the theory.

Definition 5.3 (Group order). The *order* of a group G is the number of elements contained in G .

5.2 Some immediate consequences of the group axioms

Before any further investigation into the various examples of groups that exist in mathematics we can immediately draw some consequences from the group axioms.

Theorem 5.1. *The identity elements of a group is unique.*

Proof. Suppose that e and f are two identity elements of a group $(G, *)$. Consider the group element $e * f$. Since e is an identity we know that

$$e * f = f.$$

However f is also an identity and so

$$e * f = e.$$

Immediately then we can draw the conclusion that

$$e = f.$$

And so the identity element in a group is unique. $\square$

Theorem 5.2. *The inverse of a group element is unique.*

Proof. Let $(G, *)$ be a group and let $g \in G$. Suppose that the elements $f, h \in G$ are both inverses of g , i.e.

$$g * f = f * g = e \text{ and } g * h = h * g = e,$$

where e is the identity in $(G, *)$. To show that $f = h$ we shall consider the element $f * g * h$, making use of the associativity of $*$. Note that

$$f = f * e = f * (g * h) = (f * g) * h = e * h = h,$$

as required. $\square$

Theorem 5.3. *The cancellation law holds, i.e.*

$$\forall a, b, c \in G \quad a * c = b * c \Rightarrow a = b,$$

and

$$\forall a, b, c \in G \quad c * a = c * b \Rightarrow a = b.$$

Proof. This is proved immediately by multiplying both sides of the given equation by the element c^{-1} , on the right or left as appropriate. $\square$

The next result should remind you of similar results in the context of matrices and functions. It is a generalisation of those results to any group.

Theorem 5.4. *The inverse of a product is the product of the inverses, but in the opposite order, i.e.*

$$\forall a, b \in G \quad (a * b)^{-1} = b^{-1} * a^{-1}.$$

Proof. This is proved using the associativity property and the definition of inverses,

$$\begin{aligned} (a * b) * (b^{-1} * a^{-1}) &= a * (b * b^{-1}) * a^{-1}, \text{ by associativity} \\ &= a * e * a^{-1} \\ &= a * a^{-1} \\ &= e. \end{aligned}$$

And similarly one can show that

$$(b^{-1} * a^{-1}) * (a * b) = e.$$

Therefore $(a * b)^{-1} = (b^{-1} * a^{-1})$. $\square$

Theorem 5.5 (General associativity). *In any group $(G, *)$ general associativity holds. General associativity says that in any product of n elements $a_1, a_2, \dots, a_n$ the product is well-defined no matter how parentheses are inserted into the product to indicate how it should be evaluated by forming pair-wise products, e.g.*

$$((a_1 * (a_2 * a_3)) * a_4) = (a_1 * (a_2 * (a_3 * a_4))) = \text{etc} \quad (5.1)$$

As we have said above there are many examples of groups in the mathematics that we have studied so far. We now describe some important classes of groups, some using quite familiar concepts and some that may be new to you.

5.3 Number groups

We can form many groups out of the familiar number systems and operations. Remember that $\mathbb{Z}$ denotes the set of integers, $\mathbb{Q}$ denotes the set of rational numbers and $\mathbb{R}$ denotes the set of real numbers.

Example 5.1. The following pairs all form groups

$$(\mathbb{Z}, +), (\mathbb{Q}, +), (\mathbb{R}, +), (\mathbb{Q} - \{0\}, \times), (\mathbb{R} - \{0\}, \times).$$

For the addition operation the identity element is always the number 0 and the inverse of an element x is more traditionally written as $-x$. For multiplication the identity is the number 1 and the inverse of x will be the reciprocal $\frac{1}{x}$. Note that it is necessary to exclude 0 when using the operation of multiplication. This is because zero has no multiplicative inverse.

However not all such pairings will form a group.

Example 5.2. Why does the pair $(\mathbb{Z}, \times)$ not form a group ?

These examples of course are all infinite groups - the underlying sets in each case are infinite. But there also exist families of finite groups closely related to the integers and the operations of addition and multiplication.

5.4 Modular arithmetic groups

Let $n \in \mathbb{Z}$ such that $n > 1$. We can define operations on $\mathbb{Z}$ called *addition modulo n* and *multiplication modulo n* as follows. For integers a, b their sum modulo n is defined to be the integer c , which is the non-negative remainder left upon division of the integer $a + b$ by the modulus n . So we take the integers a and b and add them in the usual way, then we divide this sum by the integer n and take the non-negative remainder c ($0 \leq c < n$). This number c is the *sum of a and b modulo n* and we write

$$a + b = c \pmod{n}.$$

Modular multiplication is defined in a similar way. For integers a, b their product modulo n is defined to be the integer c , which is the non-negative remainder left upon division of the integer ab by the modulus n . So we take the integers a and b and multiply them in the usual way, then we divide this product by the integer n and take the non-negative remainder c ($0 \leq c < n$). This number c is the *product of a and b modulo n* and we write

$$ab = c \pmod{n}.$$

Definition 5.4. The group $(\mathbb{Z}_n, +)$ is the group formed by the integers

$$0, 1, 2, 3 \dots n - 1$$

under the operation of addition modulo n . It is called the *additive group modulo n* .

The definition of a group using multiplication modulo n is not so straightforward. The issue is the existence of multiplicative inverses modulo n . An integer a will only have an inverse for the operation of multiplication modulo n if and only if it is *co-prime* to n , i.e. if a and n share no common factors apart from 1.

Example 5.3. 1. The multiplicative inverse of 2 modulo 5 is 3 as

$$2 \times 3 = 6 = 1 \pmod{5}.$$

2. The multiplicative inverse of 3 modulo 8 is itself as

$$3 \times 3 = 9 = 1 \pmod{8}.$$

3. The element 2 does not have a multiplicative inverse modulo 8, as 2 is not co-prime to 8.

4. The multiplicative inverse of 5 modulo 11 is 9 as

$$5 \times 9 = 45 = 4 \times 11 + 1 = 1 \pmod{11}.$$

Definition 5.5. We define the set $\mathbb{Z}_n^\times$ to consist of those integers between 1 and n that are co-prime to n , i.e.

$$\mathbb{Z}_n^\times = \{a \in \mathbb{Z} : 1 \leq a \leq n, a \text{ is co-prime to } n\}.$$

Example 5.4. Determine the elements contained in the following

1. $\mathbb{Z}_4^\times = \{\dots\}$
2. $\mathbb{Z}_5^\times = \{\dots\}$
3. $\mathbb{Z}_6^\times = \{\dots\}$
4. $\mathbb{Z}_7^\times = \{\dots\}$
5. $\mathbb{Z}_8^\times = \{\dots\}$

From these examples can you describe the elements contained in $\mathbb{Z}_p^\times$, where p is a prime number, i.e. a number with no positive integer factors other than itself and 1.

Definition 5.6. The pair $(\mathbb{Z}_n^\times, \times)$ where the operation $\times$ is multiplication modulo n forms a group. It is called the *multiplicative group modulo n* .

5.5 Cayley tables

Finite groups, such as these modular arithmetic groups, can be studied by way of their *Cayley table*.

Definition 5.7 (Cayley table). A *Cayley table* for a finite group $(G, *)$ is a two-dimensional table, whose rows and columns are indexed by the elements of G . The entries in the body of the table show the results of combining the row and column index elements with the operation $*$. See the generic Cayley table in figure 5.1. Take note of the convention regarding the ordering of the factors in the product $g_i * g_j$, the row index element is on the left and the column index element is on the right. It is important to have this convention and to stick to it as not all groups are Abelian.

$*$	$\dots$	g_j	$\dots$
$\vdots$			
g_i	$\dots$	$g_i * g_j$	$\dots$
$\vdots$			

Figure 5.1: A generic Cayley table

Example 5.5. We show some Cayley tables for some modular arithmetic groups.

1. The table for the group $(\mathbb{Z}_4, +)$ is

$+$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

$(\mathbb{Z}_4, +)$

2. The table for the group $(\mathbb{Z}_5^\times, \times)$ is

$\times$	1	2	3	4
1	1	2	3	4
2	2	4	1	3
3	3	1	4	2
4	4	3	2	1

$(\mathbb{Z}_5^\times, \times)$

3. The table for the group $(\mathbb{Z}_8^\times, \times)$ is

$\times$	1	3	5	7
1	1	3	5	7
3	3	1	7	5
5	5	7	1	3
7	7	5	3	1

$(\mathbb{Z}_8^\times, \times)$

In the exercises you are asked to provide the Cayley tables for some other modular arithmetic groups. Note that, because they are based on the usual arithmetic operations of addition and multiplication, all of the modular arithmetic groups are Abelian, i.e. both types of operation are always commutative.

The study of modular arithmetic is an important aspect of number theory, and this topic and some of its group-theoretic aspects will be explored further in the Level 5 unit Number Theory and Cryptography.

5.6 Cyclic groups and generators

Before continuing with examples of groups we give some more definitions and introduce some more group theory.

Definition 5.8 (Multiplicative notation). Let g and h be elements of a group $(G, *)$. The group product $g * h$ can also be denoted gh . In this way we often omit the use of a separate symbol for the group operation and simply use this *multiplicative notation* for the group products as long as the actual operation being used is clearly understood from the context. Similarly the notation g^n , where n is a positive integer denotes the product

$$g^n = g * g * \cdots * g,$$

where there are n factors g in the product. When the power is negative it denotes products of the inverse element of g , i.e. if $m > 0$ then

$$g^{(-m)} = g^{-1} * g^{-1} * \cdots * g^{-1},$$

where there are m factors g^{-1} in the product. Lastly, the notation g^0 denotes the identity element of the group.

Definition 5.9 (Cyclic group). A group G is *cyclic* if there is an element $g \in G$ such that

$$G = \{g^n : n \in \mathbb{Z}\}.$$

The element g is called a *generator* of G and we use the notation

$$G = \langle g \rangle.$$

A cyclic group may have more than one generator. In the exercises you are asked to explore modular arithmetic groups and search for examples of cyclic groups amongst them. You should find there that generators are not unique, i.e. a cyclic group G can have more than one generator. Indeed if g is a generator of G then so will g^{-1} be (although g^{-1} need not necessarily be a different element to g).

Definition 5.10 (Cyclic subgroup, subgroup). If we take any element g of a group G , then the set

$$\langle g \rangle = \{g^n : n \in \mathbb{Z}\},$$

will form a group under the same operation used in G . This is called a *cyclic subgroup* of G *generated by* g . More generally a *subgroup* of a group G is a group H under the same operation as that used in G such that $H \subset G$.

Definition 5.11 (Order of an element). Let g be an element of a group G . The order of the cyclic subgroup $\langle g \rangle$ is also referred to as the *order of the element* g .

The order of a group element g can be characterised as the smallest positive integer n such that $g^n = e$, the identity element of the group.

5.7 Matrix groups

We can form matrix groups using either the operation of matrix addition or matrix multiplication. We need to take care to distinguish the different uses of the words identity and inverses. These might be used in their matrix-theoretic sense or their group-theoretic sense. Sometimes the meanings will coincide and sometimes not. It should be clear from the context what is meant in each case.

Definition 5.12. The group $(M(n, \mathbb{R}), +)$ is the group formed by all $n \times n$ matrices with real entries, under the operation of matrix addition.

The identity in this group is the $n \times n$ zero matrix. The inverse of an element A is the matrix $-A$.

Similar groups can also be formed by restricting the matrix entries to come from the other number systems $\mathbb{Z}, \mathbb{Q}, \mathbb{C}$. Or even more generally, we could choose any group as the source of the entries.

For the operation of matrix multiplication the identity element will always be the appropriately sized identity matrix (in the linear algebra sense) since multiplication by an identity matrix leaves the other matrix unchanged. The inverse elements (in the group theoretic sense) will be the inverse matrices (in the linear algebra sense). This leads us to the following definition

Definition 5.13 (General linear group). The *General Linear Group* $\text{GL}(n, \mathbb{R})$ is the group formed by the set of non-singular matrices (i.e. with non-zero determinant) of size $n \times n$ under the operation of matrix multiplication. The identity element of this group is the $n \times n$ identity matrix I_n and the inverse group element of a matrix A is the inverse matrix A^{-1} .

Recall the multiplicative property of the determinant, namely

$$\det(AB) = \det(A)\det(B),$$

for $n \times n$ matrices A and B . This property ensures that the closure axiom is satisfied for the following matrix group.

Definition 5.14 (Special linear group). The *Special Linear Group* $\text{SL}(n, \mathbb{R})$ is the group formed by the set of matrices with determinant equal to 1, of size $n \times n$ under the operation of matrix multiplication. The identity element of this group is the $n \times n$ identity matrix I_n and the inverse group element of a matrix A is the inverse matrix A^{-1} .

Example 5.6. Prove that $\text{SL}(n, \mathbb{R})$ forms a group.

See the exercises for examples of other more specialised matrix groups.

5.8 Permutation groups

Definition 5.15 (Permutation). A *permutation* of the n objects

$$1, 2, 3, \dots, n$$

is a bijective function with domain and co-domain both equal to $\{1, 2, 3, \dots, n\}$.

Definition 5.16 (Symmetric group). The *symmetric group* S_n is the group formed by all the permutations of $1, 2, \dots, n$ under the operation of composition.

Theorem 5.6. The group S_n has order $n!$.

Notation for permutations

There are two main types of notation used for permutations: the *two-line* notation and the *cycle* notation.

Definition 5.17 (Two-line notation). We illustrate this with the example

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix}.$$

The top row lists the arguments of the permutation σ and the bottom row shows the corresponding values taken by σ on those arguments, i.e.

$$\sigma(1) = 2, \sigma(2) = 4, \sigma(3) = 1, \sigma(4) = 3.$$

Example 5.7. Remembering that function compositions are read from the right, find the two-line notation for the following products of permutations. Also the convention is that, although these products are exactly compositions of functions (the permutations), we usually omit the function composition symbol $\circ$ and simply concatenate the two-line notations to denote the product.

1. In S_3 ,

$$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}.$$

2. In S_8 ,

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 4 & 5 & 1 & 6 & 8 & 2 & 7 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 8 & 4 & 6 & 1 & 3 & 5 & 7 & 2 \end{pmatrix}.$$

Definition 5.18 (Cycle notation). A *cycle* is a permutation that *cyclically permutes* a selection of the objects $1, 2, \dots, n$. For example the cycle $\sigma \in S_8$ denoted by the following *cycle notation*

$$\sigma = (1\,3\,5\,2\,4),$$

is the permutation satisfying

$$\sigma(1) = 3, \sigma(3) = 5, \sigma(5) = 2, \sigma(2) = 4, \sigma(4) = 1,$$

and where the objects not mentioned in the cycle notation are left fixed by the permutation, i.e.

$$\sigma(6) = 6, \sigma(7) = 7, \sigma(8) = 8.$$

A nice way of representing permutations is to express them as products of *disjoint* cycles.

Definition 5.19 (Disjoint cycles). Two cycles $\sigma, \tau \in S_n$ are *disjoint* if the objects that are not left fixed by σ are all different to the objects not left fixed by τ .

Example 5.8. Find expressions for the permutations of the previous example as products of disjoint cycles.

Example 5.9. Find disjoint cycle product expressions for the following products of cycles,

1. $(13)(245)(235)$
2. $(123)(13)(45)$
3. $(246)(12)$

Another nice way of representing permutations is to express them as products of cycles of length two. Such cycles are called *transpositions*.

Definition 5.20 (Transposition). A *transposition* $\sigma \in S_n$ is a cycle of length 2.

Theorem 5.7. *Any permutation $\sigma \in S_n$ can be expressed as a product of transpositions.*

Proof. First find an expression for σ as a product of disjoint cycles. Then express each cycle as a product of transpositions using the following result,

$$(a_1 a_2 a_3 \dots a_m) = (a_1 a_2)(a_2 a_3)(a_3 a_4) \dots (a_{m-2} a_{m-1})(a_{m-1} a_m).$$

□

Example 5.10. Express the following permutations from S_5 as products of transpositions,

1.

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 2 & 1 \end{pmatrix}$$

2.

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 3 & 2 & 5 & 4 \end{pmatrix}$$

The significance of transpositions is given by the following result.

Theorem 5.8. *When a given permutation σ is expressed as a product of transpositions, the number of such transpositions will always be even, or always be odd.*

This is called the *parity* of the permutation σ . Any permutation has either even or odd parity.

Theorem 5.9. *The set of all even permutations from the group S_n is denoted A_n , and it forms a group of order $\frac{n!}{2}$.*

Definition 5.21 (Alternating group). The group A_n is called the *Alternating group* of degree n .

Exercises 5.1. These problems require the application of various concepts and results from this chapter. We shall use these as a basis for discussion in the tutorials.

1. Construct Cayley tables for a selection of the modular arithmetic groups $(\mathbb{Z}_n, +)$ and $(\mathbb{Z}_n^\times, \times)$, for $1 \leq n \leq 20$.
2. For the groups you have examined in the previous question, identify which ones are cyclic and which ones require more than one generator.
3. Write down in two-line notation all the permutations from the group S_3 .
4. Write down in two-line notation all the permutations from the group S_4 .
5. In the group S_8 determine the two-line notation form of the following products or permutations,

a)

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 4 & 7 & 5 & 6 & 8 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 3 & 4 & 5 & 6 & 7 & 8 & 1 \end{pmatrix}$$

b)

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 4 & 6 & 8 & 1 & 3 & 5 & 7 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 8 & 4 & 6 & 1 & 3 & 5 & 7 & 2 \end{pmatrix}$$

6. Now express the products you found in the previous question as products of disjoint cycles.
7. Express the following permutation products as products of disjoint cycles,
 - a) $(135)(24)(36)$,
 - b) $(2467)(354)(62)$,
 - c) $(13)(25)(23)(147)(365)$.
8. Express the following permutations as products of transpositions and determine the parity of each one,
 - a) $\sigma = (1357)(24)$,
 - b) $\tau = (2573)(64)$,
 - c) $\pi = (1578369)$
9. In the group S_5 , find an element τ such that $\sigma\tau = (15)(24)$, where $\sigma = (13)(45)$.
10. In the group S_9 find the permutations σ and τ that solve the following equations,
 - a) $\sigma(1379) = (12)(45)$,
 - b) $(13)(57)\tau(134) = (1369)$.
11. Are any of the symmetric groups S_n cyclic? Determine those that are and those that are not.
12. Are any of the symmetric groups S_n Abelian? Determine those that are and those that are not.

5.9 Going further: Subgroups and Lagrange's theorem

Group theory is a huge subject in pure mathematics. Aside from applications to other areas of mathematics, one of the main concerns of group theory is classifying and enumerating all the existing groups, both finite and infinite.

To give a small taste of this subject we shall state and prove Lagrange's theorem, which places restrictions on the sizes of subgroups that might exist within a larger group.

Definition 5.22 (Subgroup). Let $(G, *)$ be a group and H a subset of G . We say that H forms a *subgroup* of $(G, *)$ if the pair $(H, *)$ is a group. Note that it is the same operation $*$ involved in the containing group G and the subgroup H . This is important as in many contexts there is more than one group like operation.

Theorem 5.10 (Lagrange's theorem). *Suppose that H forms a subgroup of a finite group G . Then the order of H divides the order of G .*